

An analysis of strand and horizontal density distributions in oriented strand board (OSB)

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Oriented strand board (OSB) is a well-established wood based panel product. The production and consumption has strongly increased during the past decade. It is well known that most of the wood based panel properties are strongly related to the specific gravity of the panel. The EN 300 requires that the specific gravity within a panel or the horizontal density distribution (HDD) is within $\pm 10\%$. For identification of different influences of raw material and process parameters, computer simulation programs of HDD were developed under laboratory conditions (Dai, Steiner 1994a, b, c). In this study commercial strands and OSB panels are investigated to verify and improve the suitability of the computer simulation models for industrial conditions. First, a wide range of strand dimensions is determined where strand width seems to exhibit the greatest coefficient of variation. Second, an analysis of the density within an OSB panel of 8 ft by 24 ft indicates that HDD clearly exceeds the EN 300 limit. Finally, the computer simulation program was improved by considering the effects of strand types (dimensions and shape), amount of fines, strand orientation, and compartment of strands due to orienting discs and rollers during mat forming. Results of the improved computer simulation correspond accurately to the experimental HDD. The model predicts that the fines and the strand compartment strongly affect the panel density distribution.

Analyse der Strands und der horizontalen Dichteverteilung von OSB

OSB ist ein etablierter Werkstoff, dessen Produktion während der letzten 10 Jahre stark angestiegen ist. Die Eigenschaften sind bekanntlich eng mit der Dichte korreliert. Die EN 300 fordert für die horizontale Dichteverteilung (HDD) eine Toleranz von $\pm 10\%$. Zur Identifizierung verschiedener Rohstoffe und Prozeßparameter auf die HDD, wurde zuvor im Labor ein Simulationsprogramm entwickelt. Mit der vorliegenden Untersuchung soll nun-

mehr, daß der Simulation zugrundeliegende Modell an kommerziellen Platten verifiziert und für Industriebedingungen angepaßt werden. Zunächst wurden die Dimensionen einer Vielzahl von Strands untersucht, wobei die Strandbreite die größten Variationen aufwies. Danach ergab eine Dichteanalyse an OSB-Platten ($2,22 \times 7,32$ m), daß die HDD deutlich die EN-Forderung übersteigt. Das Computerprogramm wurde verbessert, indem die Art der Strands (Dimensionen, Form) und deren Feinanteil sowie die Strand-Orientierung und Kompartimentierung durch Scheiben und Fächerwalzen während der Mattenformung berücksichtigt wurden. Die so erzielten Ergebnisse entsprechen nahezu exakt den experimentell gewonnenen Daten. Demnach beeinflussen der Feinanteil und die Kompartimentierung in besonderem Maße die Dichteverteilung.

1 Introduction

A substantial increase in production and consumption of oriented strand board (OSB) has taken place in various parts of the world over the past two decades. In 1997 the production in North America reached 14 mio m^3/a while in Europe a production of ca. 1 mio m^3/a was achieved (FESYP 1998). The establishment of European standards has provided the support for consumption of OSB in Europe, especially for structural applications. In comparison to the North American standards (e.g. CSA), the European standard generally calls for high product quality of OSB (EN 300).

It is well known that most of the panel properties are strongly related to the specific gravity (SG) of panel (Kollmann and Côté 1968). Therefore, the variation of specific gravity within a panel or horizontal density distribution (HDD) has to be as small as possible under the viewpoints of an efficient process control and raw material cost reduction. Furthermore, EN-standard requires a maximum variation of $\pm 10\%$ of SG within a panel (EN 300). This allows small factors of safety to be used in engineering calculations for structural applications. Additionally, reduction of HDD variation is important with respect to process equipment, because mat density variations affect the distribution of mat pressure requested for consolidation during hot pressing (Dai and Steiner 1994a). Normally, greater density variations result in higher pressure variation and maximum pressure. This subsequently requires thick press belts for continuous presses and/or additional equipment to soften the mat before

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hot-pressing (e.g. microwave). Moreover, mat density significantly influences the heat and moisture transfer during hot pressing via conductivity and permeability (Humphrey, Bolton 1989; Haas et al. 1998) which affects curing conditions and viscoelastic behavior of the mat and the panel (densification and spring back) (Suchsland 1973; Bolton et al. 1989).

To minimize the variation of HDD, it is necessary to understand the influence and significance of raw material and process parameters. Previous investigations showed that particle/strand dimensions strongly influence HDD and panel properties (Suchsland 1962; Kusian 1968; Walter et al. 1979; Xu, Steiner 1995). For quantitative understanding of the problem, a computer model of mat formation was recently developed (Dai, Steiner 1994a, b, c). The model, which was based on physical and mathematical theories, predicted mat density variations during forming and pressing as a function of strand dimensions and configurations. However, so far the model has only been verified under laboratory conditions.

The objectives of the study were to: 1) compare the model simulations with data obtained directly under industrial conditions of the OSB manufacturing process, 2) improve the model prediction accuracy based on the analysis results of geometric characteristics of commercial strands and the HDD of industrial OSB panels, and 3) discuss the options for improvement of process control and equipment.

2

Current simulation of the horizontal density distribution

Producing wood-based panels with particles, fibers, flakes or strands results in variations of density in both the vertical direction (across the panel thickness) and the horizontal direction (parallel and perpendicular to production flow). While the vertical density profile is mainly influenced by the unsteady physical-mechanical properties of the mat and curing conditions during hot-pressing (Bolton et al. 1989; Humphrey, Bolton 1989; Wolcott et al. 1990), the horizontal density distribution is mainly controlled by the geometric properties of strands/particles and the mat forming process (Neusser et al. 1972; Suchsland, Xu 1989; Dai, Steiner 1994a; Xu, Steiner 1995). In general, the horizontal density variation increases with average panel density and strand dimensions of strands (Suchsland 1959; Kusian 1968; Arnold 1974; Dai, Steiner 1995; Xu, Steiner 1995).

Dai and Steiner (1994a, b, c) presented a mathematical-physical model describing the structure of an OSB-mat by means of geometric probabilities. Input parameters for the computer simulation are characteristics of strands (length,

width and thickness) and of panel (thickness, ratio of compression and specimen size). Through different functions (binomial, exponential and Poisson) the distributions of strand organization in a single layer of mat is defined. This information is used for further calculations (e.g. area of strand overlap and void volume) and for simulating a virtual panel by combining several single layers. Based on the weight per area of mat, the density of the resultant panel can be calculated when considering the panel thickness. To define the horizontal density distribution, the standard deviation is used.

3

Investigations on commercial strands

3.1

Material and methods

Commercial dry strands were gathered from an OSB mill. Dust and fines were separated from strands using a screen. The length and the width of 1850 strands were measured using an image analyzer. The black and white pictures had 512×480 pixels with a solution of 8 bit. For determination of strand width and length of the irregular strands, polygon approximation with a maximum of eight edges was made (Fig. 1). The thickness of 125 strands was measured manually with a micrometer.

Based on the polygon, the strand shape was determined as well. Eight types of shape were defined. Each strand was classified by using the software program "Object Analyzer" (Hubert 1996).

3.2

Results and discussion

In general, a wide variation of strand dimensions was observed. The average of the strand length is approximately 60 mm (Table 1). The distribution of the length, which appears to peak at both 60 mm and 110 mm, is not a normal distribution. The 110 mm peak is related to the target which is set by the distance of the scoring knives. The frequent length of 60 mm seems to be related to the flaking process as well but is not clearly explainable.

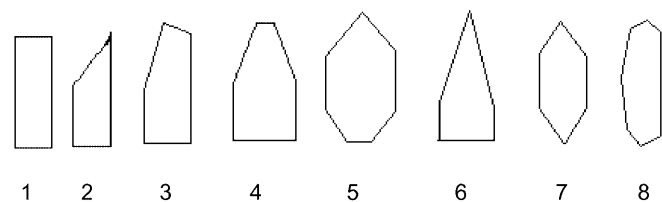


Fig. 1. Defined strand types with different shapes

Bild 1. Definierte Typen von OSB-Strands verschiedener Formen

Table 1. Dimensions and form factors of the analyzed strands

Tabelle 1. Dimensionen und Formfaktoren der analysierten Strands

1850 samples	Length [mm]	Width [mm]	Thickness [mm]	Area [mm ²]	Length/width [°]	Length/thick. [°]
Average	59.8	15.6	0.63	1019	5.1	94.9
Standard deviation S	28.7	10.0	0.23	927	3.8	45.5
Coeff. of variation V [%]	48	64	37	91	75	48
Max	117.9	73.8	1.7	6192	33.8	187.2
Min	10.7	2.1	0.2	30	0.5	17.1

Strand width and thickness are nearly normally distributed. The former appears to have the highest variation with a variation coefficient of 64%. During flaking, strand width is more resultant of other set parameters than controlled. The parameters influencing the width distribution may include feeding and cutting speed, knife angles, wood moisture and temperature conditions and target strand thickness. On the other hand, the strand thickness is usually controlled by the knife gap and feeding speed. The final strand dimensions depend upon processes after flaking because strands are bound to be broken due to excessive handling (May 1978).

An interrelation between strand thickness and width is such that width increases with increasing thickness (Fig. 2) (also see Geimer 1988). This is due to the fact that thicker strands are stiffer and therefore less prone to breaking during handling.

After separation and classification, the majority (61%) of strands are of rectangular or nearly rectangular shape (type 1, 2, 3) (Table 2). The analysis of slenderness ratio and aspect ratio indicates that strand length is on average 5 to 7 times higher than width, whereas strand thickness is mostly 100 times less than length.

The image analysis is proved to be effective in this study for evaluation of strand length and width. Similar techniques were used in other studies to measure particle and fiber dimensions, especially length and width, whereas measurement of thickness is more difficult (Arnold 1986; Plinke 1987; Spaven et al. 1993). As will be shown in the following sections, strand dimensions have a strong effect on horizontal density distribution and understandably other properties of OSB as well. Therefore, optimization of panel properties requires reliable equipment and continuous control of strand dimensions.

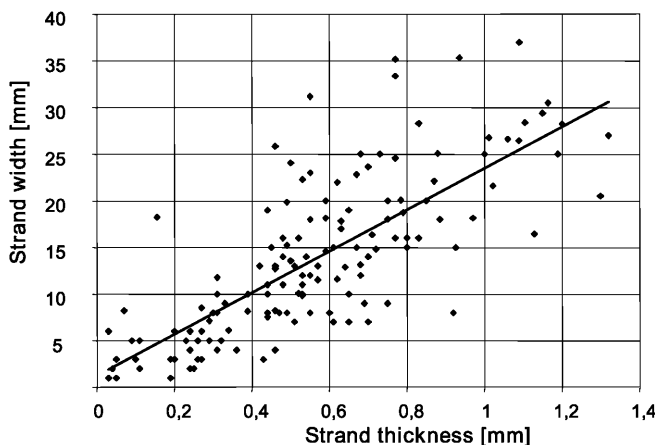


Fig. 2. Interrelation between strand thickness and width
Bild 2. Zusammenhang zwischen Dicke und Breite von Strands

Table 2. Dimension of different strand shapes
Tabelle 2. Dimensionen verschiedener Strand-Typen

Average	Type 1	Type 2	Type 3	Type 4	Type 5	Type 6	Type 7	Type 8
Frequency [%]	30	16	15	7	5	7	7	13
Length [mm]	70.9	66.8	71.0	58.1	61.7	53.2	48.9	76.9
Width [mm]	12.5	8.4	11.1	10.1	9.4	7.9	6.7	11.9
Length/width [/]	5.7	8.0	6.4	5.8	6.6	6.7	7.3	6.5
Length/thickn. [/]	118.2	111.3	118.3	96.8	102.8	88.7	81.5	128.2

4 Investigations of horizontal density distribution of commercial OSB panels

4.1

Material and methods

A full-size commercial OSB panel was investigated. The panel which is 8 ft (7.30 m) in width, 24 ft (2.40 m) in length and 3/8" (9.5 mm) in thickness, consists of 90% of aspen (*populus tremula*) and 10% of hardwoods (mainly alder and birch). Powder phenol-formaldehyde was used as binder. The full size panel was divided in six smaller panels (Fig. 3). These panels were stored in a defined climate (relative humidity 65%, temperature 20 °C) over 4 weeks in FORINTEK laboratories resulting in an average moisture content of 5,9%.

The panel density in sections 1 and 2 was tested gravimetrically (EN 323). Specifically, sections 1 and 2 were cut into 25 × 25 mm² specimen which resulted in a total of 3528 specimens. One hundred of selected 25 × 25 mm² specimens were further cut into 400 specimens of 11 × 11 mm². To determine the density for larger specimens, local averages were calculated based on neighboring specimens of 25 × 25 mm². Sections 3 to 6 were tested by using an industrial gamma-ray measuring unit. Diameter of the inspection area was 72.9 mm. The panels were investigated in a grid of 50 by 50 mm².

4.2

Results and discussion

A clear horizontal density variation in the industrially produced OSB was observed. In Fig. 4 the influence of the specimen size in relation to HDD is shown, i.e. HDD increases as the specimen size decreases. For a specimen

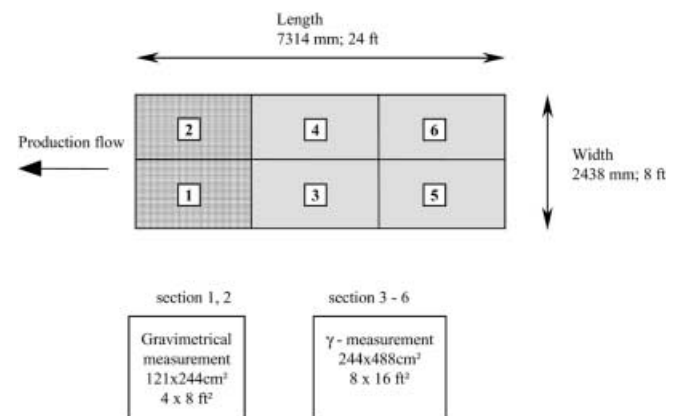


Fig. 3. Investigated commercial OSB panel
Bild 3. Untersuchte kommerzielle OSB-Platte

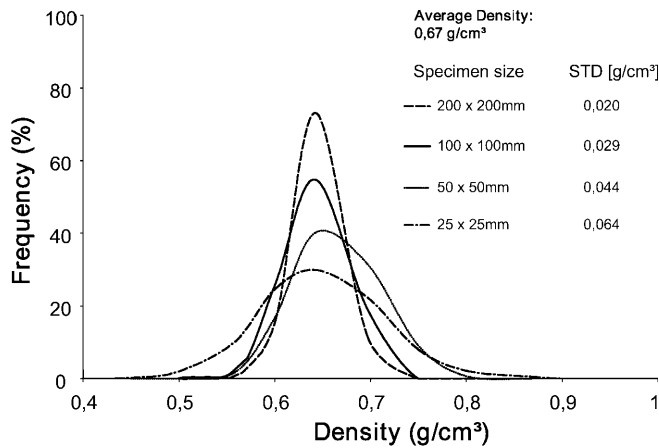


Fig. 4. Interrelation between specimen size and HDD of a commercial OSB panel (gravimetrical measurement)
Bild 4. Zusammenhang zwischen Probengröße und horizontaler Dichteverteilung der kommerziellen OSB-Platte (gravimetrische Bestimmung)

size of $50 \times 50 \text{ mm}^2$, 95% of the gravimetrically measured densities were localized in a range between 0.57 g/cm^3 to 0.77 g/cm^3 . With regard to EN 300, which requires a tolerance for the density of $\pm 10\%$, the allowable range of

variation is from 0.60 g/cm^3 to 0.74 g/cm^3 . This implies that the investigated commercial OSB panel did not fulfill the requirements of the actual EN standard.

In Fig. 5 the horizontal density distribution (γ -measurement) of the sections 3 to 6 is shown. The density of the investigated area of 2.44 m by 4.88 m varies between 0.50 g/cm^3 and 0.85 g/cm^3 . No clear pattern of the density distribution can be identified. For further analyses, the panel was virtually separated in stripes of 2.54 cm parallel to production direction. This evaluation showed an obvious distribution of density (Figs. 6, 7). During hot pressing the mat is spread which results in lower density at the panel edges. Although the panel was trimmed by 5 cm from the edge before the investigations, the reduction of density at the edge still exists (Fig. 7). In general, the mean and variation of panel density on the left side of the production is higher than those on the right side. This indicates a non-uniform mat forming process in which the variation of density may be caused by variation in bulk density in the strand bins and uneven spread of strands along the forming heads.

Previous investigations have shown a strong horizontal density variation in particleboard as well. Neusser et al. (1972) investigated a full size particleboard of lower quality. Although the boundary conditions were slightly different (thickness 19 mm , density 0.70 g/cm^3 ,

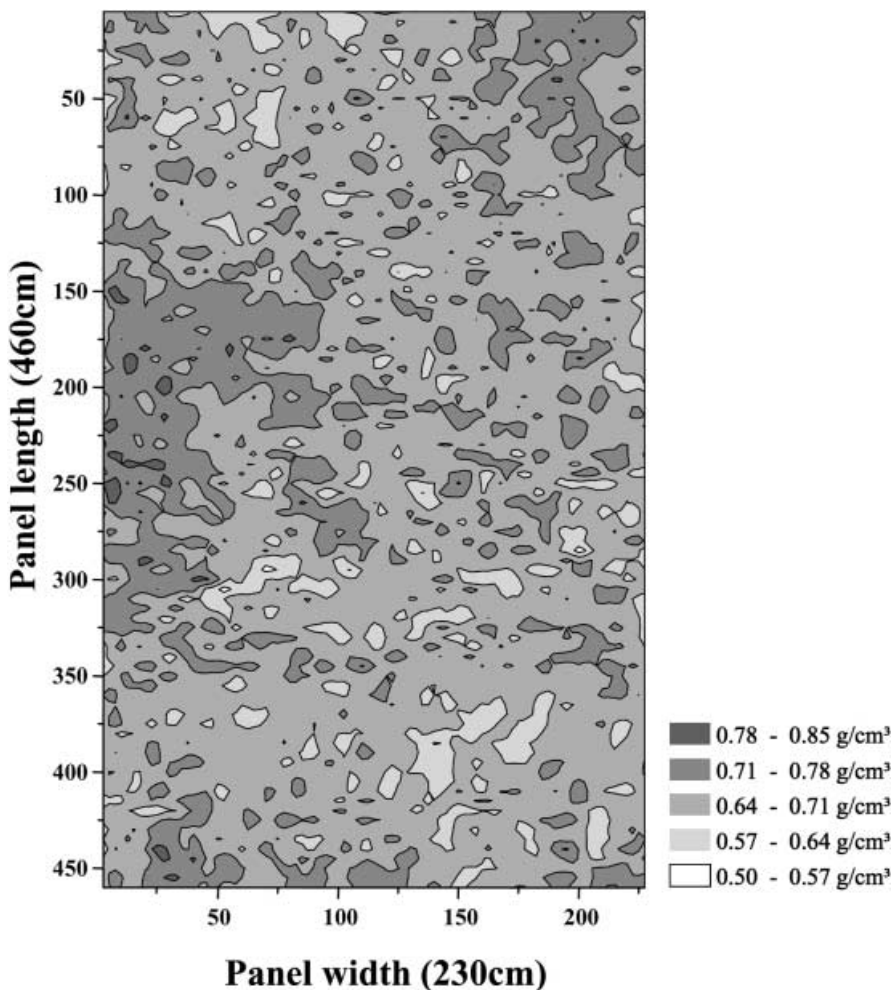


Fig. 5. Density distribution of a commercial OSB panel (γ -ray measurement)
Bild 5. Dichteverteilung der kommerziellen OSB-Platte (γ -Strahl Messung)

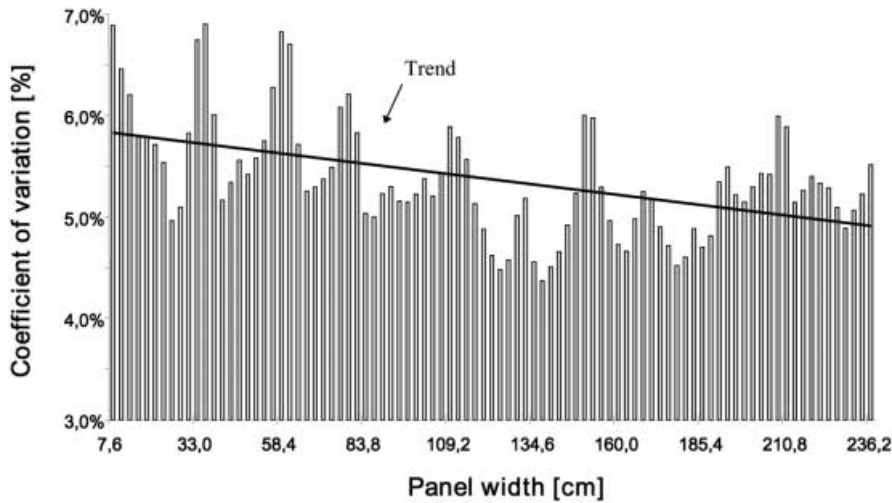


Fig. 6. Density distribution of 2.54 cm by 4.88 m stripes in production direction (coefficient of variation)

Bild 6. Dichteverteilung von Plattenstreifen (2.54 cm × 4.88 m) in Produktionsrichtung (Variationskoeffizient)

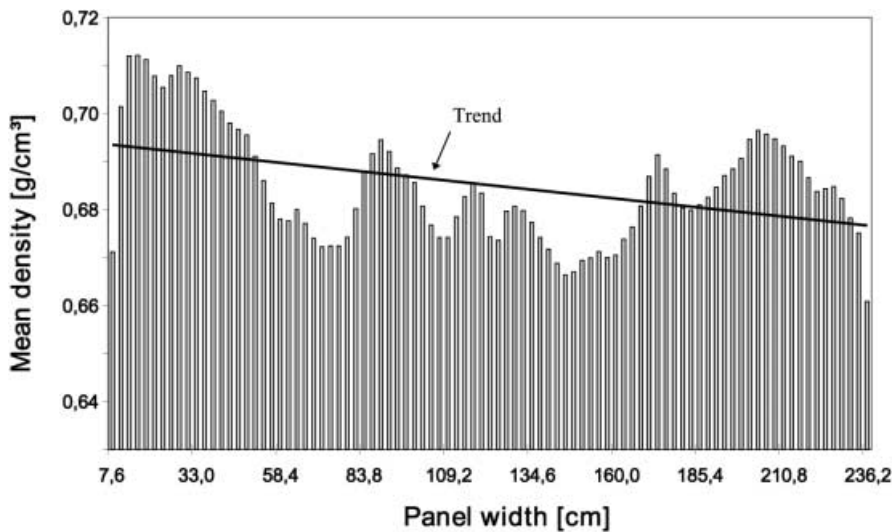


Fig. 7. Density distribution of 2.54 cm by 4.88 m stripes in production direction (average density)

Bild 7. Dichteverteilung von Plattenstreifen (2.54 cm × 4.88 m) in Produktionsrichtung (Mittelwert der Dichte)

Table 3. Comparison of density variation of different commercial panel types
Tabelle 3. Vergleich der Dichteveriation in verschiedenen kommerziellen Platten

		MDF	Particleboard structural grade	Particleboard furniture grade	Waferboard	OSB
Particle		Fibre	Particle	Particle	Wafer	Strand
Panel thickness [mm]		16.3	12	16	11.1	9.3
Density [g/cm ³]	$\bar{x}$	0.81	0.76	0.65	0.67	0.67
	V	1.0%	3.1%	2.4%	4.7%	10.0%
Source:		Böhme 1988	Gressel 1981	Gressel 1981	Xu/Steiner 1994	Current investigation

specimen size from $120 \times 120 \text{ mm}^2$ to $27 \times 27 \text{ mm}^2$, old process technology), some comparison are still be possible. The coefficient of density variation was 7.2% (specimen size: $27 \times 27 \text{ mm}^2$). In contrast, particleboard panels produced using modern technologies yield a much lower coefficient of variation (approximately 1 to 3%) (Gressel 1981; Böhme 1988). The smaller variation is mainly attributed to the use of new mat forming technology, such as the roller formers which allows for more precise control of particle distribution along and across the production direction. The newer forming

systems are also equipped for close-loop process controls of weight per area, which additionally support a more homogeneous mat forming.

In general, it can be stated that larger particle size results in increased variation of HDD. A comparison of MDF, particleboard, waferboard and OSB is shown in Table 3. MDF produced with the smallest particles (single fibers or fiber bundles) shows the smallest variation of HDD (1%). This is because smaller particles result in smaller overlapping areas, finer voids between particles and more chance for randomization. An OSB panel which

is produced with large strands, results in a high variation of HDD. Due to this, some OSB panels may fail to meet the standard or customer requirement. Therefore it is particularly important to control the distribution of mass weight per area during OSB mat forming via process control and equipment improvement.

5 Computer simulation of HDD in commercial OSB

The current computer simulation model (Dai and Steiner 1994b, c) was used to calculate the density variations or standard deviation of panel density at various specimen sizes based on measured strand dimensions ($59.8 \times 15.6 \times 0.63 \text{ mm}^3$). The predictive results were compared with the experimental measurements. The density variation depends largely upon specimen sizes (Fig. 9). The larger the specimen, the smaller the variation. However, the degree to which the variation changes decreases with specimen size. As shown in Fig. 9, when the specimen size reaches $200 \times 200 \text{ mm}^2$, the variation changes little afterwards. It is also observed from the figure that the γ -ray measurements fit well with the gravimetric data on a single curve.

While both follow the same trend, the model prediction is generally greater than the experimental observation and the discrepancy increases as the specimen size decreases. Such a discrepancy is largely attributed to the assumptions made to simplify the model development. They are:

- uniform dimensions of strands
- uniform shape of strands
- homogeneous density of strands
- randomized distribution of strands in the panel
- randomized orientation of strands

In order to improve the accuracy of the model prediction, these assumptions need to be modified to accommodate the complexities in industrial production. The additional variables, which can be incorporated into the model, include:

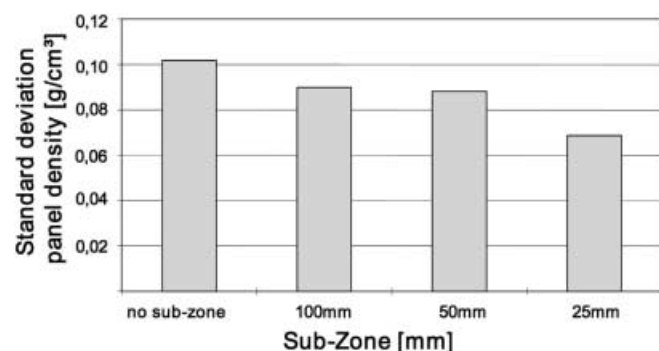


Fig. 8. Effect of sub-zones to the HDD of a simulated OSB panel (Strand dimensions: $59.80 \times 15.60 \times 0.63 \text{ mm}^3$, panel density: 0.67 g/cm^3 , panel thickness: 9.26 mm)

Bild 8. Effekt der Kompartimentierung auf die HDD einer simulierten OSB-Platte (Stranddimensionen: $59.80 \times 15.60 \times 0.63 \text{ mm}^3$, Plattendichte: 0.67 g/cm^3 , Plattendicke: 9.26 mm)

- variation of strand dimensions
- different shapes of strands
- density variations between strands due to species and locations
- compartment of strands induced by with rollers and discs at the former
- strand orientation

The strand dimensions affect the HDD in such a manner that strand thickness imposes the most profound impact followed by strand width, whereas strand length has the least effect (Dai, Steiner 1995). Such effects along with that of strand shapes can be considered in the improved computer model in which the distribution of strand dimensions and non-rectangular shapes are simulated. The use of small strands or fines reduces the porosity of the panel and results in a more regular HDD because the fines are able to fill the voids between larger strands. Similar effects are observed in particle-board (Hänsel 1988; Schweitzer, Niemz 1991; Haas et al. 1998).

During the mat forming process, strands of the surface and core layers are cross-oriented by discs and rollers to improve the directional strength properties. The orienting discs or rollers may, at the same time, cause the strands to lay randomly within the area of compartments formed by adjacent discs, whereas the number of strands distributed between the compartments may be relatively uniform. These effects can be simulated in the improved computer model by dividing the mat into sub-zones across the production width. The dimensions of the sub-zones are assumed related to the gaps and the free fall distances of the orienting discs. When operating forming heads with a sub-zone of 100 mm , the HDD can be reduced by nearly 18%. The widely used disc gaps in the OSB industry are $35\text{--}50 \text{ mm}$ which can result in a reduction of HDD by nearly 30% (Fig. 8). This is due to the fact that fewer strands are deposited at random and more strands are placed in uniform fashion, and strands are better oriented in the mat.

For simulating the HDD of the investigated commercial OSB panel, the new computer simulation was fed with the data of strands and mat composition determined as follows:

- panel size: $2.4 \times 4.8 \text{ m}^2$
- thickness: 9.3 mm
- wood density: 0.40 g/cm^3
- sub-zone size: $40 \times 40 \text{ mm}^2$
- Mass in surface layers: 25%
- Mass in core layer: a) 50% strands b) 34% strands + 16% fines
- orientation of surface layers:
 - parallel to production flow (standard deviation 20°)
- orientation of core layer:
 - perpendicular to production flow (standard deviation 30°)

Results of the new computer simulation model correspond more accurately with the experimental results (Fig. 9, Table 4). The maximum difference between simulation and experiment is only 0.003 g/cm^3 . According to

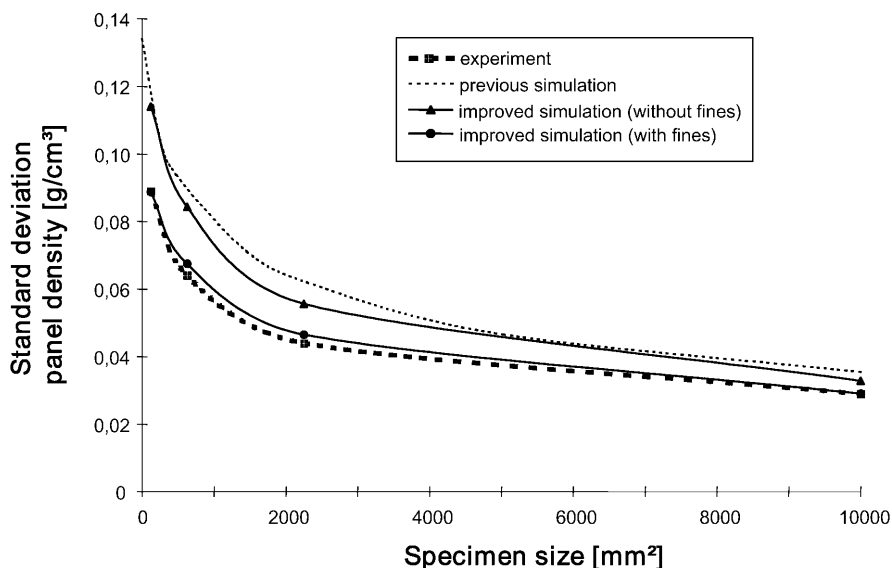


Fig. 9. Specimen size vs. standard deviation of density (commercial OSB). Comparing results of computer-simulations and experiment
Bild 9. Probengröße versus Standardabweichung der Dichte (kommerzielle OSB). Vergleichende Ergebnisse von Computersimulation und Experiment

Table 4. Standard deviation of the density of industrial OSB. Comparing results of computer-simulation and experiment (gravimetric)

Tabelle 4. Standardabweichung der Dichte von Industrieplatten. Vergleich der Computersimulation mit experimentellen (gravimetrischen) Ergebnissen

Mean density 0.67 g/cm ³ Specimen size	Experiment S [g/cm ³]	Simulation old S [g/cm ³]	Improved S [g/cm ³]	Difference simulation vs. Experiment [g/cm ³]	
				Old	Improved
11 mm × 11 mm	0.089	0.114	0.089	0.025	0.000
25 mm × 25 mm	0.064	0.088	0.067	0.024	−0.003
50 mm × 50 mm	0.044	0.061	0.047	0.017	−0.003
65 mm × 65 mm*	0.038	0.051	0.035	0.013	−0.001
100 mm × 100 mm	0.029	0.037	0.029	0.008	0.000
200 mm × 200 mm	0.020	0.020	0.017	0.000	0.003

S: Standard deviation

* (γ-measurement)

the model, the fines and the strand compartment have the strongest effects on the density distribution.

6

Conclusions

The existing computer simulation of the OSB mat forming has been improved to account for industrial conditions. Based on the results, the following conclusions can be drawn with regard to improved process control:

- Stranding parameters (i.e. machinery parameters, moisture content) can be adjusted to reach preferable strand dimensions and shape. Especially strand thickness and width should be taken into account to effectively reduce the variation of HDD and properties within panel.
- The use of fines reduces HDD but may be restricted due to high glue consumption.
- Orientation and compartment of strands with discs and rollers reduce horizontal density distribution.
- HDD characterizes the local mat density variations, which can affect mat pressure and heat and moisture transfer during hot pressing.
- The investigated OSB panels failed to fulfill the EN requirements for specific gravity. Questions therefore re-

main regarding the level of requirement and testing method (i.e. specimen size) for OSB.

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